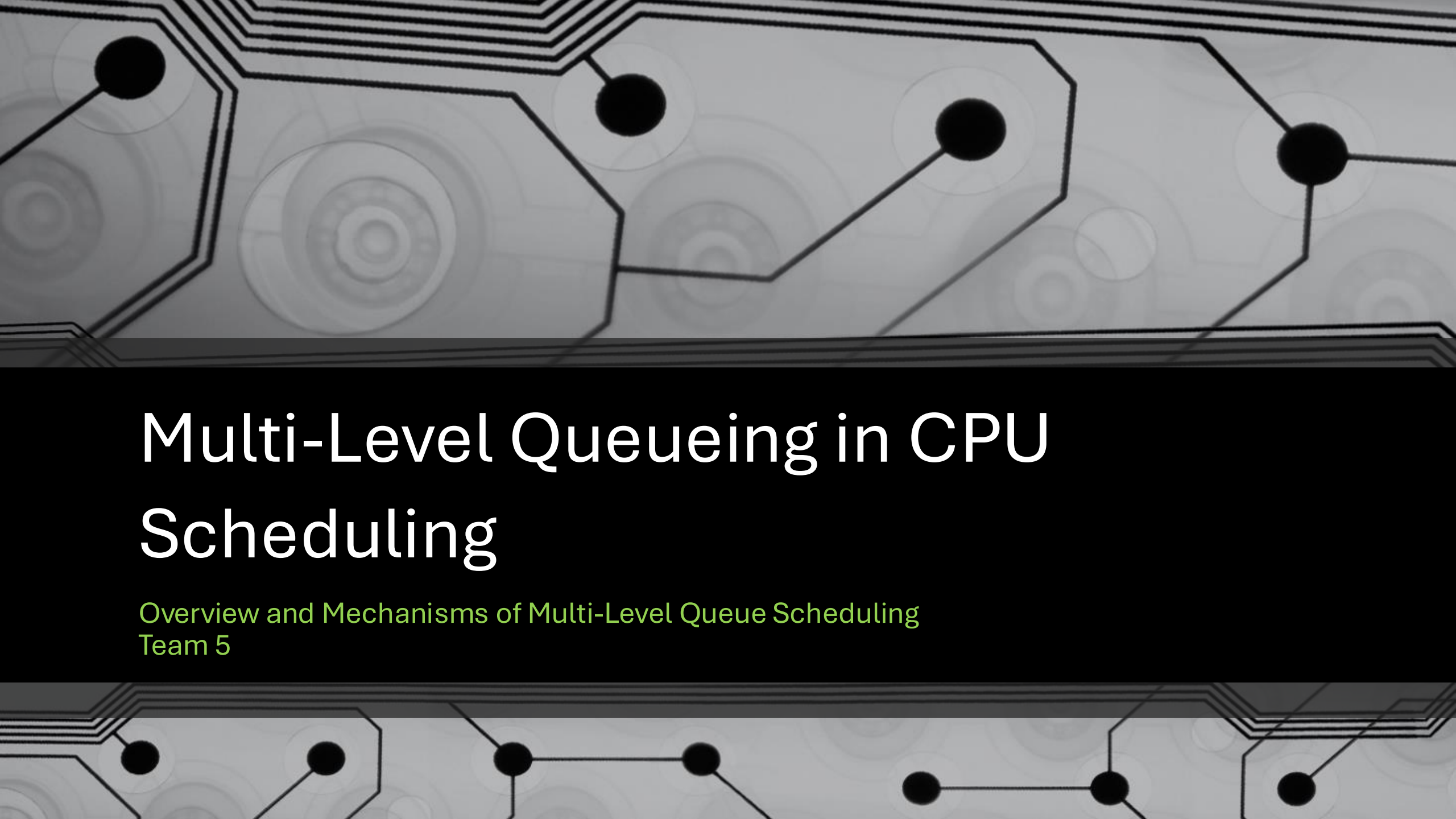
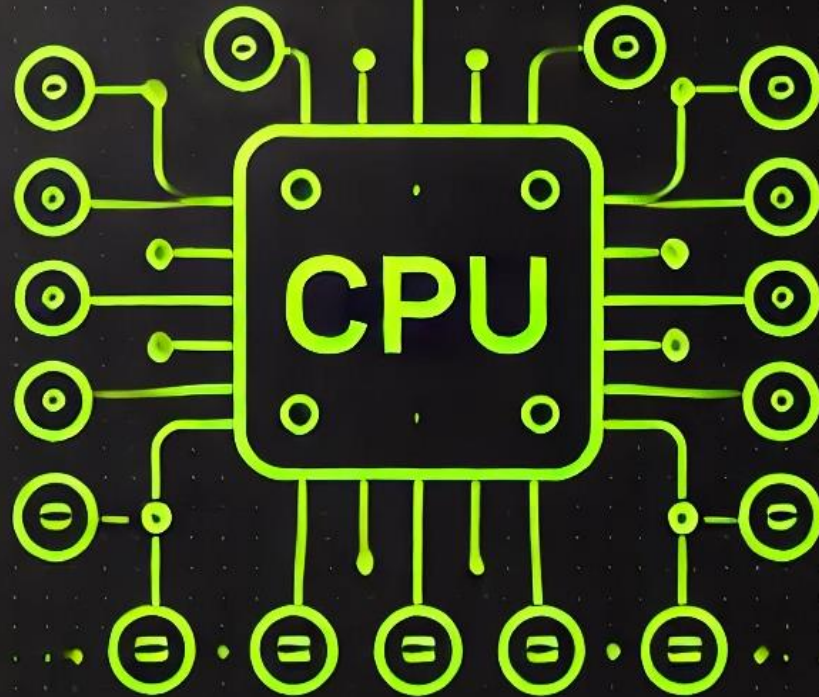




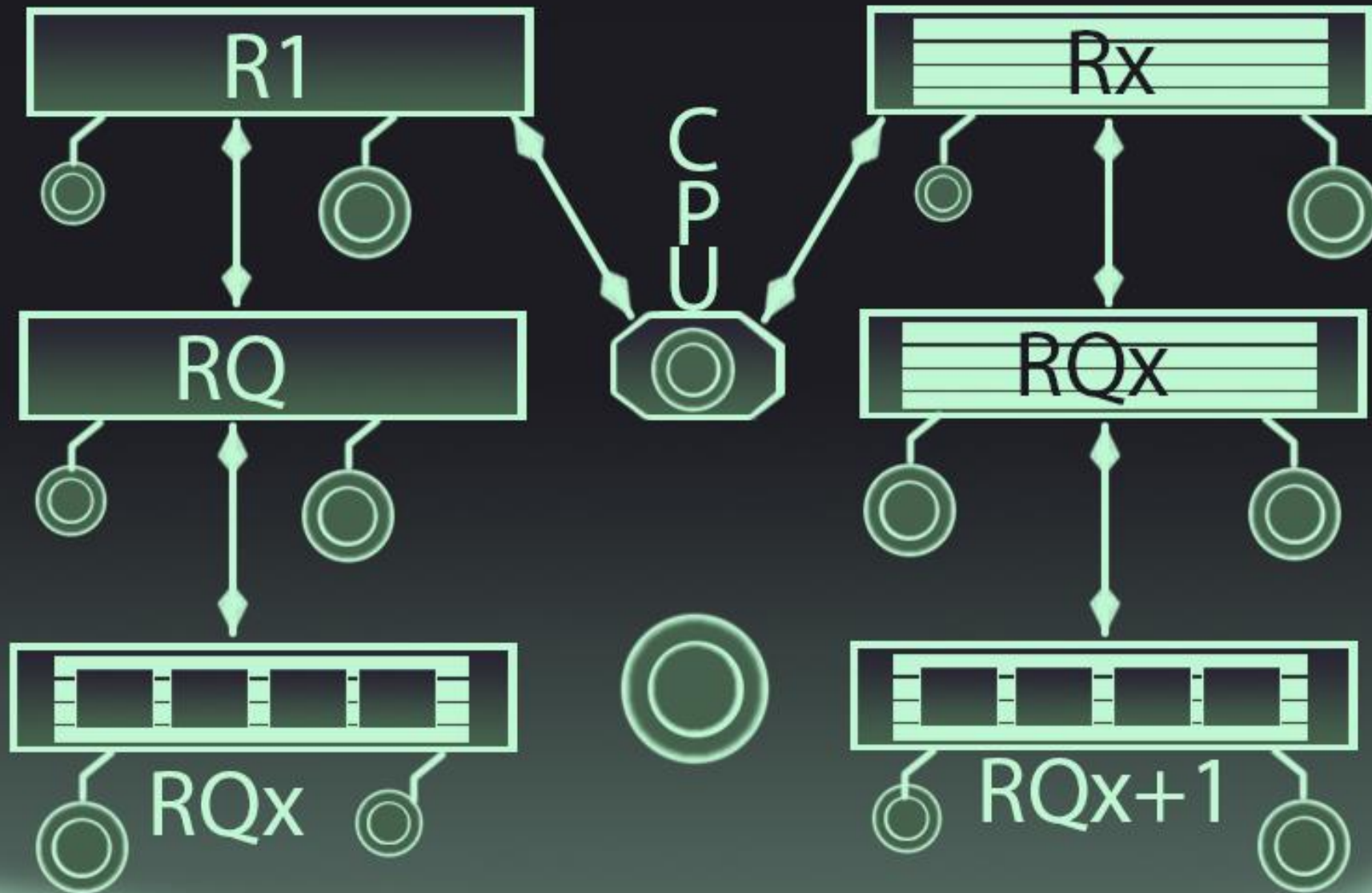
Multi-Level Queueing in CPU Scheduling

Overview and Mechanisms of Multi-Level Queue Scheduling
Team 5

INTRODUCTION TO **CPU Scheduling**

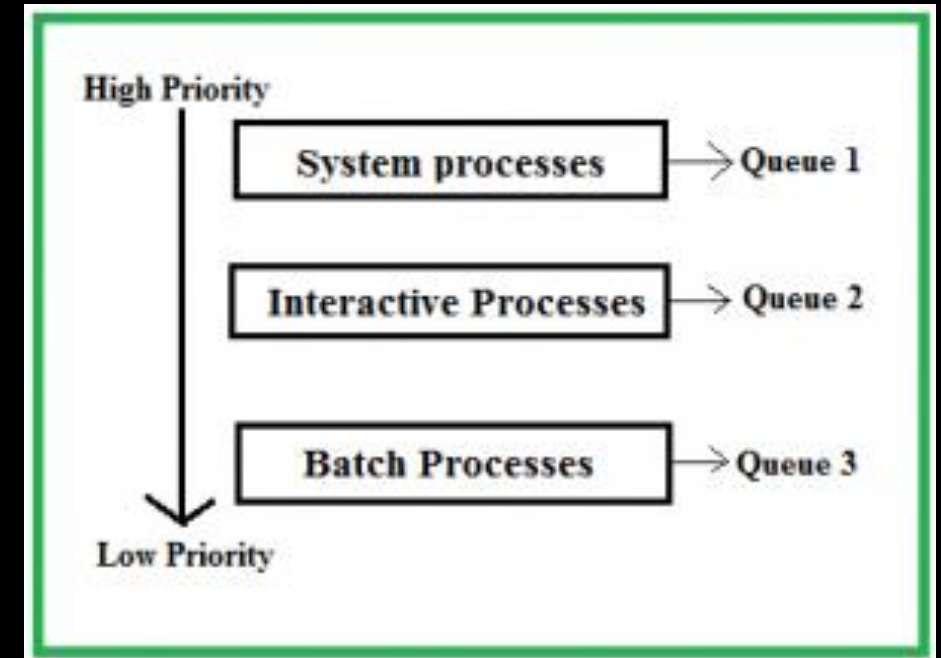


Basics of Multi-level Queueing



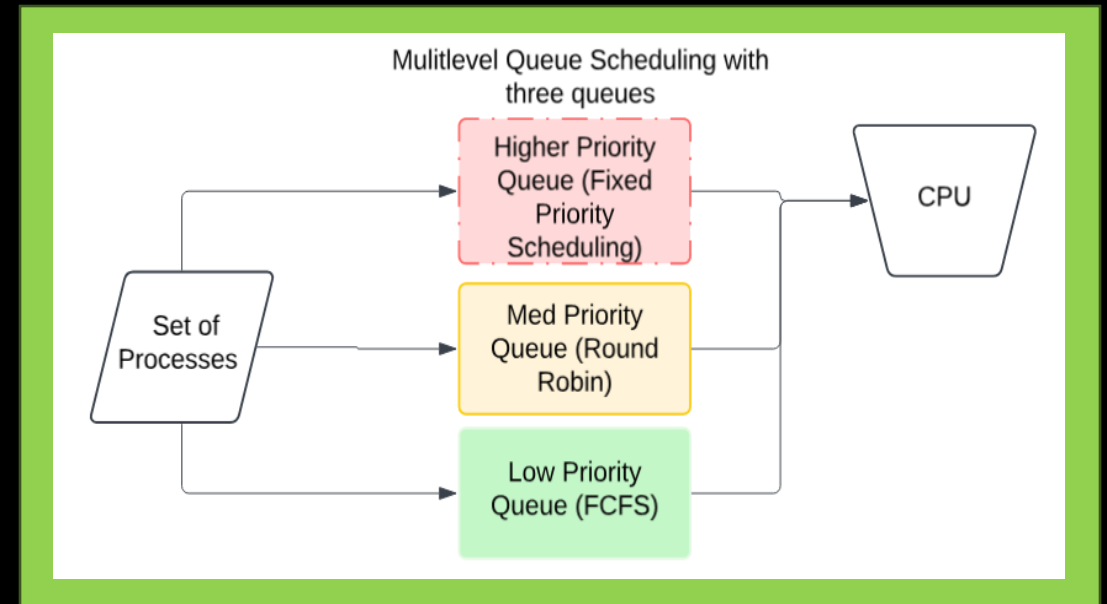
Structure of Multi-Level Queue Scheduling

- **How it Works:**
 - Processes are assigned to queues based on criteria like priority, memory needs, or CPU burst time.
 - Common queues: foreground (interactive processes), background (batch processes).
- **Example Queue Types:**
 - System queue, Interactive queue, Batch queue, etc.



Types Of Scheduling within Multi-level Queues

- **Fixed Priority Scheduling:**
 - Higher-priority processes are always executed before lower ones.
 - May lead to starvation in lower-priority processes.
- **Round Robin Scheduling:**
 - ◻ Often used within individual queues, especially for time-sharing systems.
 - ◻ Helps reduce starvation by giving each process a time slice.
- **First Come First Serve:**
 - Background queues typically use FCFS because they are not interactive and lack urgency
- **Can use any other scheduling algorithm**



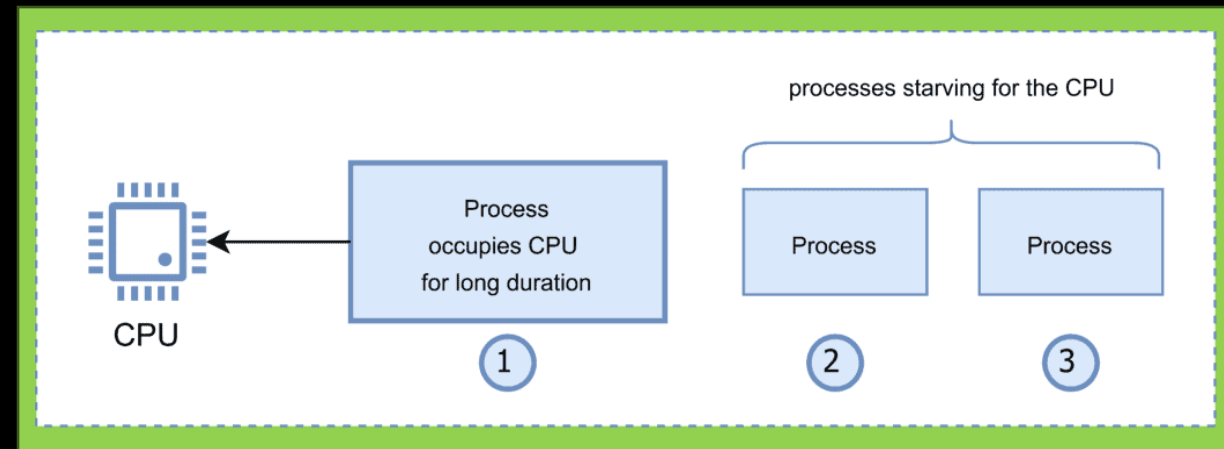
Advantages of Multi-level Queue Scheduling

- **Efficiency:**
 - Segments processes, reducing overall wait times.
- **Simplicity:**
 - Clear organization of processes based on similar attributes.
- **Customization:**
 - Flexibility to apply different scheduling algorithms per queue.



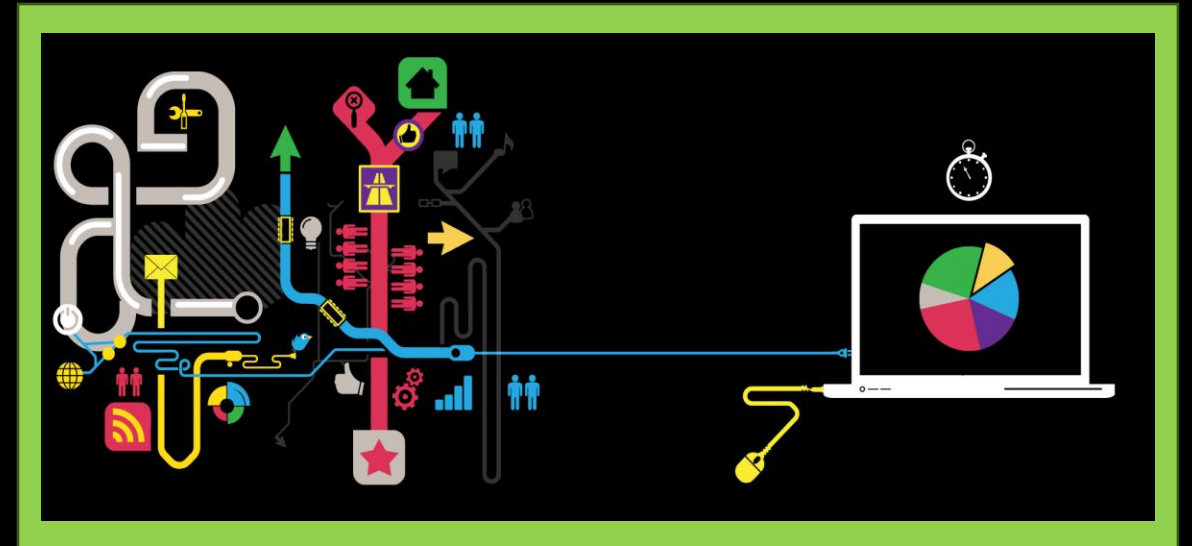
Disadvantages of Multi-Level Queue Scheduling

- **Starvation:**
 - Lower-priority queues may starve if higher-priority queues dominate CPU time.
- **Complexity in Management:**
 - Requires careful design to avoid process imbalance.
- **Limited Flexibility:**
 - Once a process is assigned to a queue, it remains there, even if its needs change.



Applications of Multi-Level Queue Scheduling

- **Real-time Systems:**
 - Ensures critical tasks are processed first.
- **Interactive Systems:**
 - Prioritizes user interaction and real-time responses.
- **Batch Processing Systems:**
 - Separates batch processes for background execution.



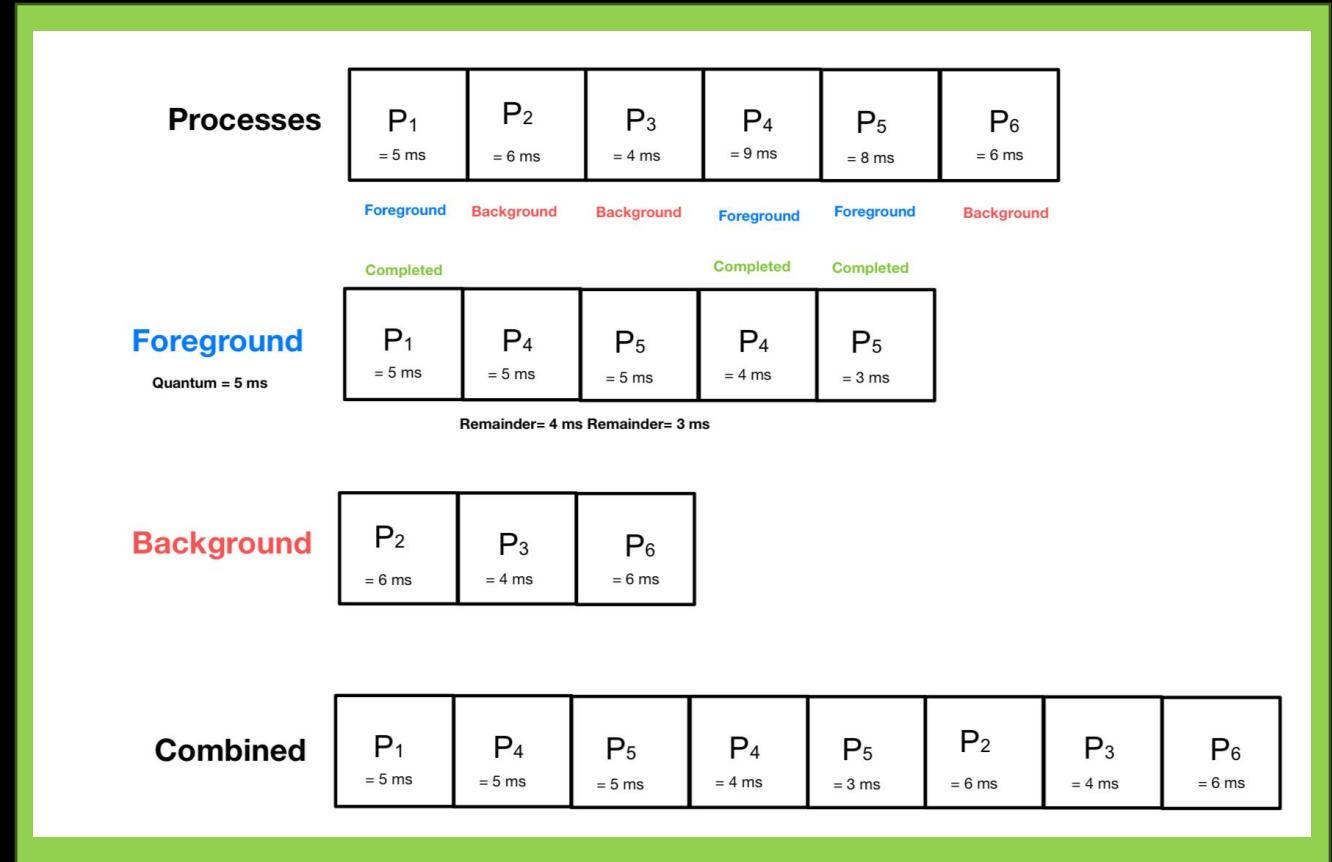
Example of Multi-Level Queue Scheduling

- **Scenario:**

- Processes arrive all at $t = 0$ ms
- Foreground: Uses Round Robin algorithm
- Background: Uses FCFS algorithm
- Quantum for foreground is 5 ms

- **Results**

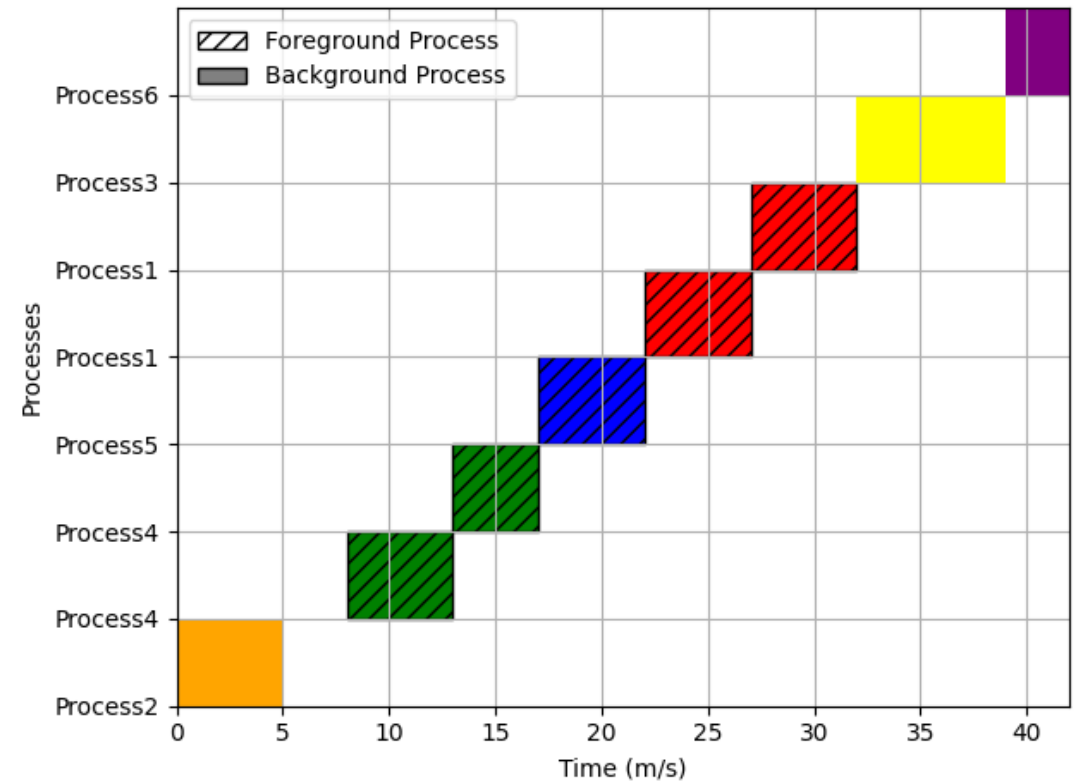
- Notice that all foreground processes complete before background processes
- CPU utilization 100% when processes arrive at $t = 0$ ms



Multi-Level Queue Scheduling Code Example

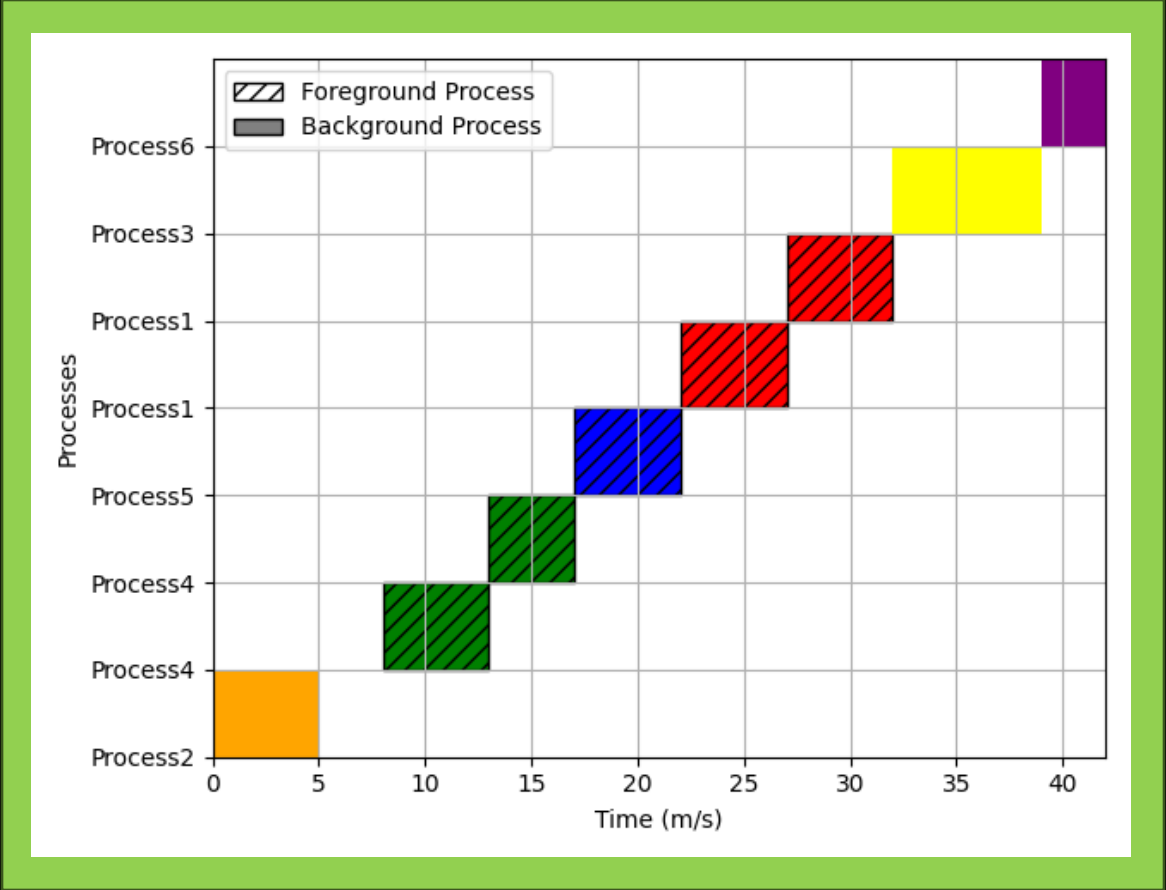
Foreground queue uses Round Robin with a time quantum of 5 milliseconds. Below is the process table.

Process	Type	Burst	Arrival
1	Foreground	10	15
2	Background	5	0
3	Background	7	8
4	Foreground	9	8
5	Foreground	5	13
6	Background	6	10



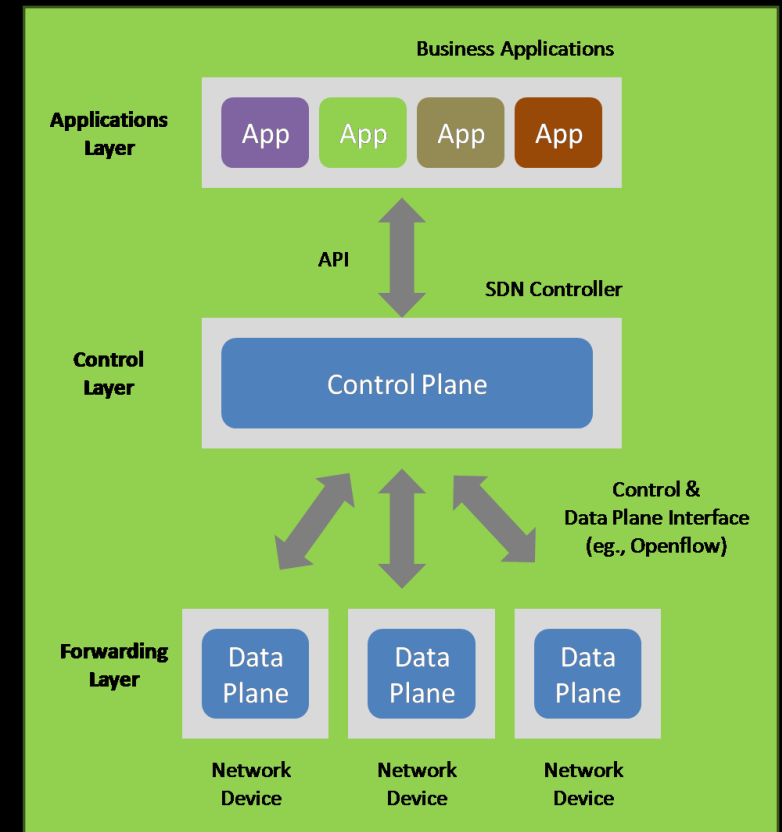
Multi-Level Queue Scheduling Code Example Continued

Calculations	
Total Completion Time	45 ms
Idle Time	3 ms
Avg. Waiting Time	10.667 ms
Avg. Turn Around Time	17.667 ms
Throughput	0.1333 processes per ms
CPU Utilization	93.33%



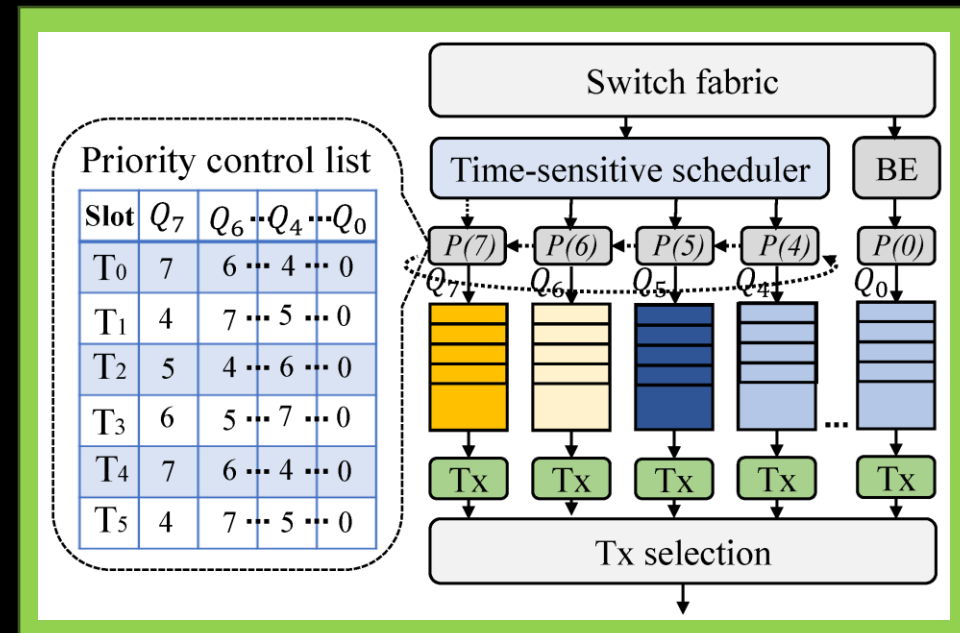
Software Defined Networking and MLQ

- Software defined networking improves upon the traditional multi-layer networking architecture
- Breaks down the layers into software components
- SDN controller acts as an intermediary
- Proposes a model for packet priority
- MLQ with packets
- Improvements in packet queue size, bits/second, and packet loss



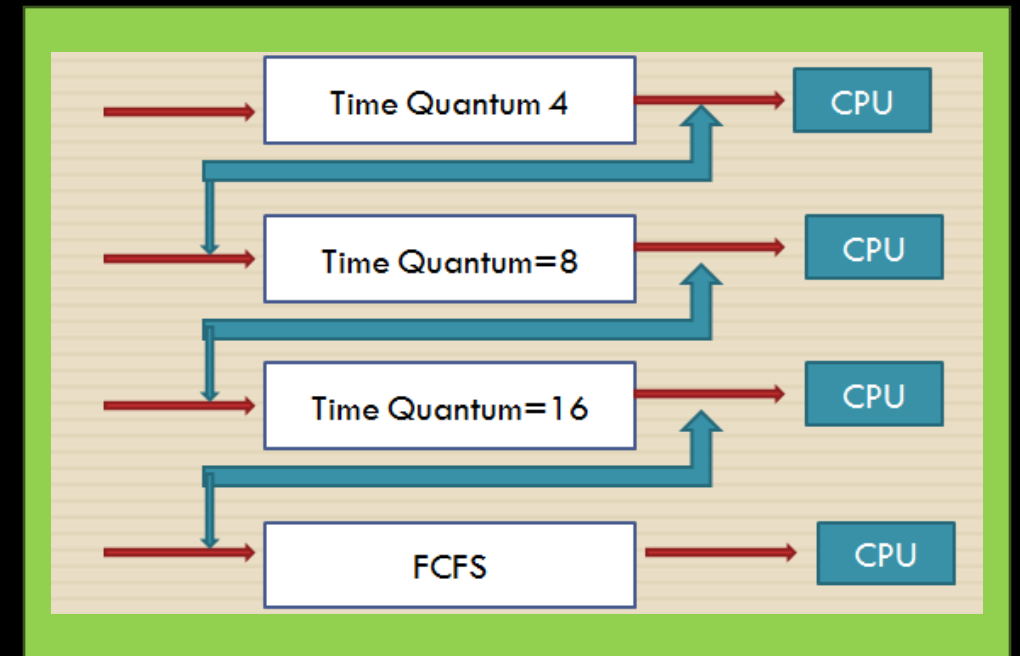
LEO Satellites and MLQ

- Low Earth orbit satellites need to be improved so that time sensitive flows (TSFs) can be efficiently handled
- Using MLQ within the time-sensitive network (TSN) technology helps overcome the challenge of the large space-time range and limited resources
- Simulation evaluations show a 90% and 14.5% reduction in packet loss and time-out ratio, respectively
- Mechanism using MLQ can handle more time-sensitive flows



Multilevel Feedback Queue Scheduling

- A bit different from the naïve multilevel queue scheduling
- Allows processes to move into different queues based on a defined system (demotion & promotion)
- Very flexible
- Helps fix the starvation issue associated with multilevel queue scheduling
- Good for dynamic, real-time environments



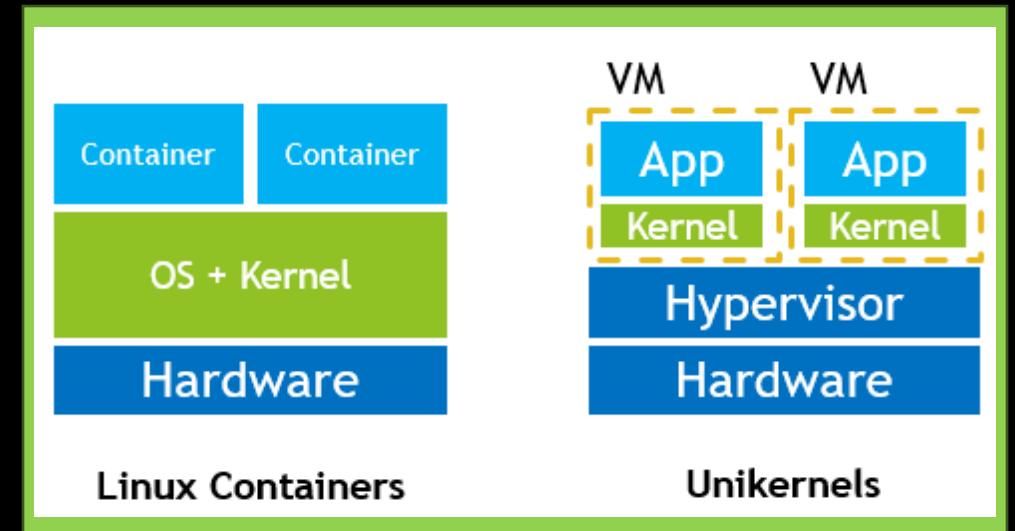
Multilevel Feedback Queue Scheduling Technique for Healthcare Systems

- **Scenario:**

- Overcrowding in healthcare facilities leads to long wait times and decreased quality of care.
- Multilevel Feedback Queue Scheduling Technique (MLFQST) to optimize resource allocation and minimize risks.
- Patients are assigned to different queues based on **urgency**.
- Dynamic feedback adjusts patient priorities over time.
- Uses First-Come-First-Served (FCFS) scheduling within queues.
- Average waiting time: Reduced to **12.9 minutes** per patient.
- Stations demonstrated **high efficiency** under the MLFQST model.

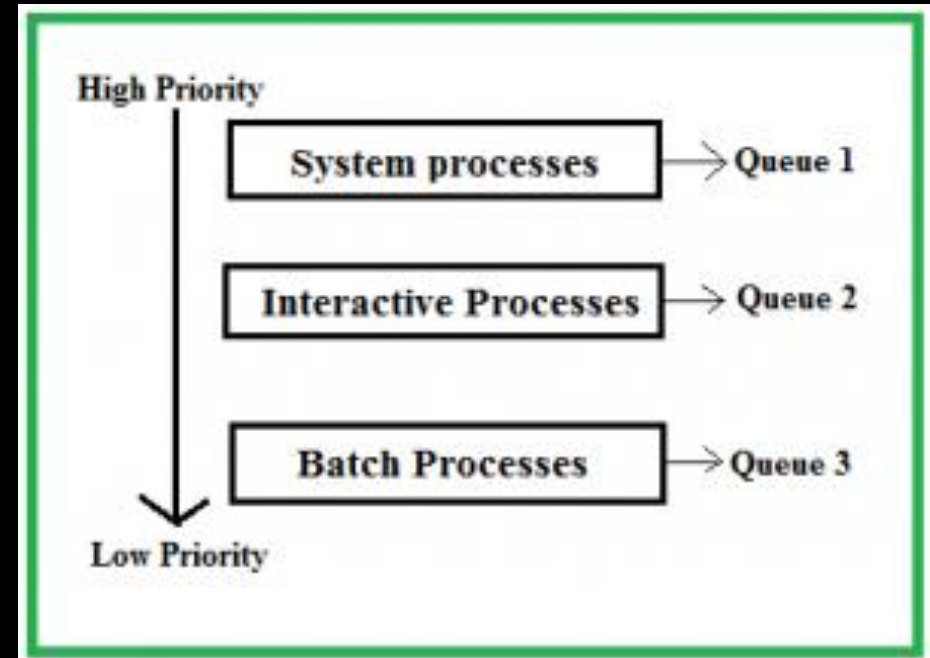
Cloud Computing and MLFQ

- Many cloud operations manage dynamic workloads
 - small-lived applications to long-running jobs
- These operations will struggle to efficiently adapt to changes in workloads priorities
- Cloud applications' operations can be made more efficient with the implementation of MLFQs
- Example:
 - Unikernels are defined by their minimal resource usage along with fast start up times. By implementing a MLFQ, the average waiting and turnaround time can be reduced by up to 15% during large-scale task scheduling



Conclusion

- CPU Scheduling Problem Solved through MLQ
- Benefits
 - Prioritization of important processes
 - Very flexible and simple implementations
- Disadvantages
 - Starvation possible
 - Pitfalls of scheduling algorithms within queues
- Improvement with MLFQ
- Applicable to real world issues



Thank you :)

References

Bonan, D., Qiusong, Y., & Mingshu, L. (2024, May 1). *M-RSF: A Multilevel Feedback Queue task scheduling mechanism for Unikernel* – Doaj. Tongxin xuebao. <https://doaj.org/article/5d380f7c3b544a33991ad3bdc4f4689e>

Harki, N. (2024). Multi-Level Feedback Queue Scheduling Technique. *Cihan University-Erbil Scientific Journal*, 8(2), 36-42. <https://doi.org/10.24086/cuesj.v8n2y2024.pp36-42>

Ma, X., Li, S., Guan, Z., Li, J., Sun, H., Wang, Y., & Guo, H. (2023). Time-Sensitive Networking Mechanism Aided by Multilevel Cyclic Queues in LEO Satellite Networks. *Electronics* [Basel], 12(6), NA. <http://dx.doi.org.aurarialibrary.idm.oclc.org/10.3390/electronics12061357>

Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating Systems Concepts* (10th ed.). Wiley.+

Umadevi, K. S., Pranay, M. S., & Rachana, K. (2017). Multilevel queue scheduling in Software Defined Networks. *2017 Innovations in Power and Advanced Computing Technologies (i-PACT)*, 1–4. <https://doi.org/10.1109/ipact.2017.8245144>